

№ 12 Четырехгранная кость

$$n = 2 \quad \lambda = 0,2$$

$$\{1, 2, 3, 4\}$$

$$H_0: P_{01} = \frac{1}{4}, P_{02} = \frac{1}{4}, P_{03} = \frac{4}{24}, P_{04} = \frac{8}{24}$$

$$H_1: P_{11} = P_{12} = P_{13} = P_{14} = \frac{1}{4}$$

	1	2	3	4
P_0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{4}{24}$	$\frac{8}{24}$
P_1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$$L = \frac{L_1}{L_0} = \frac{P_1(x_1) P_1(x_2)}{P_0(x_1) P_0(x_2)}$$


ℓ !

$x_2 \backslash x_1$	1	2	3	4
1	1	1	$\frac{3}{2}$	$\frac{3}{4}$
2	1	1	$\frac{3}{2}$	$\frac{3}{4}$
3	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{9}{4}$	$\frac{9}{8}$
4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{9}{8}$	$\frac{9}{16}$

G

$G: \ell \geq C$

H_0

$x_2 \backslash x_1$	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
3	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{36}$	$\frac{1}{18}$
4	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{9}$

H_1

$x_2 \backslash x_1$	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

Нужно найти C , чтобы $P(\ell \geq C | H_0) \leq 0.2$

Подберем наименьший $C = \frac{3}{2} = 7$

$\Rightarrow P(\ell \geq C | H_0) = \alpha_1 = \frac{1}{36} + 4 \cdot \frac{1}{24} = \frac{5}{36}$

$P(\ell < C | H_1) = (16 - 5) \cdot \frac{1}{16} = \frac{11}{16} = \alpha_2$

$\beta = 1 - \alpha_2 = \frac{5}{16}$